Online Test 3 (30%)

- Due 31 Jul at 23:59
- Points 100
- Questions 36
- Available until 31 Jul at 23:59
- Time limit 130 Minutes

Instructions

Due Date: Week 12 Lab Session

Weighting: 30%

About this test

1. This is an online test worth 30% of the total marks for this unit.
2. It consists of 34 questions on theories and applications. Using the lab environment in the quiz is essential to find answers.
3. You may **only attempt this test once**.
4. **Do not navigate away from the test (close the browser tag) before you have completed it and submitted it**.
5. You have **120 minutes** to complete the test; exceeding this will result in a loss of marks and penalties.
6. This is a **close-book test**.
7. Once you begin the quiz, a summary will appear in the upper right corner, showing the number of questions (completed and total) and the time remaining.
8. Please bring your student ID to the test so the tutor can verify your identity.

This quiz was locked 31 Jul at 23:59.

Attempt history

	Attempt	Time	Score
LATEST	Attempt 1	129 minutes	88.92 out of 100

Correct answers are hidden.

Score for this quiz: 88.92 out of 100
Submitted 31 Jul at 12:40
This attempt took 129 minutes.
Partial

Question 1

1.25 / 2.5 pts

Which of the following statement(s) is(are) correct?

- [ ] Kafka file storage length is fixed.
- [ ] Kafka has a low-latency characteristic. Hence, it is not fault-tolerant.
- [ ] Kafka is a suitable tool for batch processes but not real-time processes.
- [x] Kafka is a suitable tool for the microservice structure and for accelerating the process to real-time operation.

Question 2

2.5 / 2.5 pts

Which of the following statement(s) is(are) correct?

i) You can use the command below in Apache Hive to find the system time and date.
hive> !date;

ii) The command below creates a folder on HDFS via Apache Hive.
hive> dfs -mkdir myfolder;

iii) The command below lists the current property values of the Hive environment.
hive> set -a;

iv) The command below enables column headers in HiveQL query results.
hive> set hive.cli.print.header;

- [x] i) is correct.
- [ ] iv) is correct.
- [ ] iii) is correct.
- [x] ii) is correct.

Week 11 - Lecture - P18-19.

Question 3

2.5 / 2.5 pts

Which of the following description(s) is(are) correct?

- [ ] Flume allows users to extend the system structure by adding sources and sinks only to existing storage layers.
- [x] The general sinks used in Flume include files on the local file system and HDFS.
- [ ] Flume can take in general sources include data from files, system logs, but not standard output from a process.
- [x] Flume allows users to extend the system structure by adding sources and sinks to existing storage layers or data platforms.

Week 08 - Lecture - P23

Question 4
2.5 / 2.5 pts

Which of the following statement(s) is(are) correct?

i) If you want to execute a file called "myquery.hql" containing HiveQL statements from the command window, you should use the command: *hive -S myquery.hql*

ii) If you want to use HiveQL directly from the command line to select everything from "users", you can use the command: *hive -f 'SELECT * FROM users'*

iii) Apache Hive does not allow users to update individual records.

iv) Apache Hive does not allow users to delete individual records.

- [x] iii) is correct.
- [x] iv) is correct.
- [ ] ii) is correct.
- [ ] i) is correct.

Week 11 - Lecture - P12, P17.

Partial

Question 5
1.67 / 2.5 pts

Which of the following statement(s) is(are) true about Sqoop?

- [ ] Sqoop is primarily designed for real-time data processing.
- [ ] Sqoop can take data from HDFS and insert it into an existing table in an RDBMS.
- [x] Sqoop supports a WHERE clause to import just a potion of a table.
- [x] Sqoop can be used for incremental data imports.

Question 6
2.5 / 2.5 pts

Which of the following statement(s) is(are) true?

- [ ] When using inverted index in MapReduce, using the byte offset of the line and the line itself as the key and the value, respectively, is a common design.
- [x] Sorting is frequently used as a speed test for a Hadoop cluster.
- [ ] Data stored in HDFS is automatically assigned with inverted index for efficient retrieval processes.

When using indexing algorithm in MapReduce, using the byte offset of the line and the line itself as the key and the value, respectively, is a common design.

Week 07 - Lecture - P07, P09-P10.

Question 7

2.5 / 2.5 pts

Given the dataset sales.txt below

```
sales.txt
Alice    2874
Bob      3929
Alice    2354
Carlos   28214
Carlos   44256
```

What is the result of the following Pig script?

```
allsales = LOAD 'sales.txt' AS (name:chararray, sale:int);
bigsales = FILTER allsales BY sale > 10000;
DUMP bigsales;
```

```
(Carlos,28241)
(Carlos,44256)
```

- displayed in the terminal

```
(Alice,2874)
(Bob,3929)
(Alice,2354)
```

- stored in file 'bigsales.txt'

```
(Carlos,28241)
(Carlos,44256)
```

- stored in file 'bigsales.txt'

```
(Alice,2874)
(Bob,3929)
(Alice,2354)
```

- displayed in the terminal

Week 9 lecture slides, pg23, pg43

Partial

Question 8

1.25 / 2.5 pts

Given the tables customers and orders below,

customers

cust_id	name	country
1	Eileen	US
2	Chloe	AU
3	Jaycob	UK
4	Ethan	UK
5	Tom	AU

orders

order_id	cust_id	total_price
000857	1	1250
001123	2	800
003027	3	950
005566	5	566
009527	5	1500

Which of the following HiveQL script produces the following result?

cust_id	name	total_price
1	Eileen	1250
2	Chloe	800
3	Jaycob	950
4	Ethan	NULL
5	Tom	566
5	Tom	1500

- [ ]
```
SELECT c.cust_id, name, total_price
FROM customers c
FULL OUTER JOIN orders o
ON (c.cust_id=o.cust_id);
```
- [ ]
```
SELECT c.cust_id, name, total_price
FROM customers c
JOIN orders o
ON (c.cust_id=o.cust_id);
```
- [x]
```
SELECT c.cust_id, name, total_price
FROM customers c
LEFT OUTER JOIN orders o
ON (c.cust_id=o.cust_id);
```
- [ ]
```
SELECT c.cust_id, name, total_price
FROM customers c
RIGHT OUTER JOIN orders o
ON (c.cust_id=o.cust_id);
```

Question 9

2.5 / 2.5 pts

Given the dataset employees.txt below

employees.txt

```
CEO     227552
Director        182515
Director        182515
Director        184515
Manager 165117
Manager 165282
Manager 145379
Engineer        75000
Engineer        75000
Engineer        76000
```

Which of the Pig script produces the following result in terminal?

Result:

```
(CEO,{(CEO,227552)})
(Director,{(Director,184515)})
(Engineer,{(Engineer,76000)})
(Manager,{(Manager,165282)})
```

```
employees = LOAD 'employees.txt' AS (title:chararray, salary:long);
title_group = group employees BY title;
top_salary = FOREACH title_group
                {sorted = ORDER employees BY salary ASC;
                 highest_salary = LIMIT sorted 1;
                 GENERATE group, highest_salary;}

DUMP top_salary;
```

```
employees = LOAD 'employees.txt' AS (title:chararray, salary:long);
title_group = group employees BY title;
top_salary = FOREACH title_group
                {sorted = ORDER employees BY salary DESC;
                 highest_salary = LIMIT sorted 1;
                 GENERATE group, highest_salary;}

DUMP top_salary;
```

```
employees = LOAD 'employees.txt' AS (title:chararray, salary:long);
title_group = group employees BY title;
top_salary = FOREACH title_group
                {highest_salary = LIMIT sorted 1;
                 sorted = ORDER employees BY salary DESC;
                 GENERATE group, highest_salary;}

DUMP top_salary;
```

```
employees = LOAD 'employees.txt' AS (title:chararray, salary:long);
title_group = group employees BY title;
top_salary = FOREACH title_group
                {sorted = ORDER employees BY salary ASC;
                 highest_salary = LIMIT sorted 1;
                 GENERATE group, highest_salary;}

STORE top_salary INTO topSalary;
```

Week9 lecture slides, pg80-82

Question 10

2.5 / 2.5 pts

You're a data engineer working on large datasets in Hive.

You frequently need to analyze data where the customers and orders tables are joined together using a common key.

What can you do to avoid typing the same script repeatedly?

- Create a "JOIN CACHE"
- Create a "PERSISTENT JOIN"
- Create a "DATA LINK"

- ( ) Create a "VIEW"

Week 11, lecture slides pg91-92

Question 11

2.5 / 2.5 pts

Which of the following statement(s) is(are) correct?

- [x] Apache Pig execution in the interactive shell will be delayed until the output is required.
- [ ] Apache Pig can only be executed in grunt.
- [ ] Apache Pig use the statement "READ" to take the input file from the source.
- [x] If you type "pig" in the command window, you can enter the interactive shell called Grunt to work with Apache Pig.

Week 09 - Lecture - P14.

Question 12

2.5 / 2.5 pts

Which of the following is NOT a design goal of Flume?

- ( ) Extensibility.
- ( ) Reliability.
- ( ) Scalability.
- (•) Data Deduplication.

Week 8, lecture slides pg17

Question 13

2.5 / 2.5 pts

Consider a MapReduce program with three input files shown below.

File1: I am a student at SUT.

File2: I currently study for Big Data Architecture and Applications.

File3: We have developed several MapReduce programs in the unit.

Mapper emits (word, file name), eg. (I, File1), (am, File1), (a, File1), ... (I, File2), ...

Reducer emits (word, file name list), eg. (I, [File1, File2]), ...

Which of the following names best describe the algorithm?

- Word Count
- Word Co-Occurrence
- Inverted Index
- Secondary Sort

Wee7 lecture slides, pg11-12

Incorrect

Question 14

0 / 2.5 pts

Assume the tables below are stored in Hive.

oid	cid	total
34	B	22.65
35	C	37.17
37	A	42.22
38	B	40.00
39	Z	13.97
40	E	63.12
41	A	37.64

cid	name	gender	country	age
A	Alice	F	US	28
B	Bob	M	AU	32
C	Carlos	F	PL	41
D	Dieter	M	JP	56

- Result 1:

cid	name	oid	total
B	Bob	34	22.65
C	Carlos	35	37.17
A	Alice	37	42.22
B	Bob	38	40
NULL	NULL	39	13.97
NULL	NULL	40	63.12
A	Alice	41	37.64

- Result 2:

cid	name	oid	total
A	Alice	37	42.22
A	Alice	41	37.64
B	Bob	34	22.65
B	Bob	38	40
C	Carlos	35	37.17

- Result 3:

cid	name	oid	total
A	Alice	37	42.22
A	Alice	41	37.64
B	Bob	34	22.65
B	Bob	38	40
C	Carlos	35	37.17
D	Dieter	NULL	NULL

Which of the following statement(s) is(are) correct?

i) The following codes produce the output in Result 1.

SELECT c.cid, name, oid, total FROM customer c LEFT OUTER JOIN orders o ON (c.cid = o.cid);

ii) The following codes produce the output in Result 2.
SELECT c.cid, name, oid, total FROM orders o LEFT OUTER JOIN customer c ON (c.cid = o.cid);

iii) The following codes produce the output in Result 3.
SELECT c.cid, name, oid, total FROM orders o RIGHT OUTER JOIN customer c ON (c.cid = o.cid);

iv) The following codes produce the output in Result 3.

SELECT c.cid, name, oid, total FROM customer c LEFT OUTER JOIN orders o ON (c.cid = o.cid);

- [ ] iv) is correct.
- [x] ii) is correct.
- [x] iii) is correct.
- [ ] i) is correct.

Week 11 - Lecture - P51

Question 15

2.5 / 2.5 pts

Given the datasets products.txt below

```
products.txt
1          laptop  2300
2          desktop 850
3          keyboard        200
4          speaker 3000
5          Random-Access Memory     1300
6          Solid State Drives       1600
```

Which of the following Pig scripts correctly splits the data into multiple relations based on the <u>quantity_sold</u> and produces the given result?

Result:

```
(5,Random-Access Memory,1300)
(6,Solid State Drives,1600)
```

( )
```
product_info = LOAD 'products.txt' AS (prod_id:int, prod_name: chararray, quantity_sold:int);
SPLIT product_info INTO best_sellers IF quantity_sold > 2000,
                        strong_sellers IF quantity_sold > 1000 AND quantity_sold < 2000,
                        average_sellers ELSE;

DUMP strong_sellers;
```

(•)
```
product_info = LOAD 'products.txt' AS (prod_id:int, prod_name: chararray, quantity_sold:int);
SPLIT product_info INTO best_sellers IF quantity_sold > 2000,
                        strong_sellers IF quantity_sold > 1000 AND quantity_sold < 2000,
                        average_sellers IF quantity_sold < 1000;

DUMP strong_sellers;
```

```
product_info = LOAD 'products.txt' AS (prod_id:int, prod_name: chararray, quantity_sold:int);
SPLIT product_info INTO best_sellers IF quantity_sold > 2000,
                        strong_sellers IF 1000 < quantity_sold < 2000,
                        average_sellers IF quantity_sold < 1000;

DUMP strong sellers;
```

```
product_info = LOAD 'products.txt' AS (prod_id:int, prod_name: chararray, quantity_sold:int);
SPLIT product_info INTO CASE WHEN quantity_sold > 2000 THEN best_sellers,
                             WHEN 1000 < quantity_sold < 2000 THEN strong_sellers,
                             WHEN quantity_sold < 1000 THEN average_sellers;

DUMP strong_sellers;
```

Week 10, lecture slides pg24-25

Question 16

2.5 / 2.5 pts

Which of the following statement(s) about TF-IDF is(are) correct?

A. TF is the number of occurrences of the term in the document.

B. TF is the percentage of the term in a document.

C. IDF is the logarithm of (the total number of documents/number of documents containing the term)

D. IDF is the logarithm of (the number of documents containing the term/the total number of documents)

- B and C are correct.
- A and D are correct.
- Only A is correct.
- A and C are correct.

Week 7 lecture slides, pg 22

Question 17

2.5 / 2.5 pts

Which of the following statement(s) is(are) correct?

- [x] Data for Apache Hive's tables are stored on the filesystem (typically HDFS), and each table maps to a single directory.
- [ ] Apache Hive was originally developed by Cloudera for data wrangling.
- [ ] Apache Hive supports subdirectories to distinguish different tables in the same database.
- [x] Apache Hive tables are really directories of files in HDFS, which is different from RDBMS.

Week 11 - Lecture - P27, P30.

Question 18

2.5 / 2.5 pts

Given the datasets products1.txt and products2.txt below

products1.txt

```
V01        Broccoli
V02        Tomato
V03        Leek
D01        Milk
P01        Lamb Steak
```

products2.txt

```
V04        Patato
P02        Pork Mince
```

Which Pig Operator should be used in place of "OPERATOR" in the pig script below to produce the given result?

Pig script:

```
products1 = LOAD 'products1.txt' AS (prod_id:chararray, prod_name:chararray);
products2 = LOAD 'products2.txt' AS (prod_id:chararray, prod_name:chararray);
data = OPERATOR products1, products2;

DUMP data;
```

Result:

```
(V04,Patato)
(P02,Pork Mince)
(V01,Broccoli)
(V02,Tomato)
(V03,Leek)
(D01,Milk)
(P01,Lamb Steak)
```

- CROSS
- JOIN
- APPEND
- UNION

Week 10, lecture slides pg22-23

Partial

Question 19

1.25 / 2.5 pts

Consider the following two Pig scripts.

Both scripts aim to search for customers living in New York city who have made orders during a specific period of time.

Pig script 1:

```
customers = LOAD 'customers.txt' AS (cust_id:int,
                                    fname:chararray, lname:chararray,
                                    address:chararray, city:chararray,
                                    state:chararray, zipcode:chararray);
orders = LOAD 'orders.txt' AS (order_id:int, cust_id:int, order_date:chararray);

filtered_by_date = FILTER orders BY order_date MATCHES '^2013-0[234]-.*$';
filtered_by_city = FILTER customers BY city == 'New York';
joined = JOIN filtered_by_date BY cust_id, filtered_by_city BY cust_id;

DUMP joined;
```

Pig script 2:

```
customers = LOAD 'customers.txt' AS (cust_id:int,
                                    fname:chararray, lname:chararray,
                                    address:chararray, city:chararray,
                                    state:chararray, zipcode:chararray);
orders = LOAD 'orders.txt' AS (order_id:int, cust_id:int, order_date:chararray);

joined = JOIN orders BY cust_id, customers BY cust_id;
filtered_by_date = FILTER joined BY order_date MATCHES '^2013-0[234]-.*$';
filtered_by_date_city = FILTER filtered_by_date BY city == 'New York';

DUMP filtered_by_date_city;
```

Which of the following statement(s) is(are) true?

- [ ] The two scripts produce the same results.
- [ ] The two scripts are equally efficient because the order of the operations does not matter.
- [ ] The second script is more efficient because it joins the two datasets before the FILTER operation.
- [x] The first script is more efficient because it filters out unnecessary data before the JOIN operation.

Question 20

2.5 / 2.5 pts

Assume the tables below are stored in Hive.

oid	cid	total
34	B	22.65
35	C	37.17
37	A	42.22
38	B	40.00
39	Z	13.97
40	E	63.12
41	A	37.64

cid	name	gender	country	age
A	Alice	F	US	28
B	Bob	M	AU	32
C	Carlos	F	PL	41
D	Dieter	M	JP	56

- Result 1:

cid	name	oid	total
B	Bob	34	22.65
C	Carlos	35	37.17
A	Alice	37	42.22
B	Bob	38	40
NULL	NULL	39	13.97
NULL	NULL	40	63.12
A	Alice	41	37.64

- Result 2:

cid	name	oid	total
A	Alice	37	42.22
A	Alice	41	37.64
B	Bob	34	22.65
B	Bob	38	40
C	Carlos	35	37.17

Which of the following statement(s) is(are) correct?

i) The following codes produce the output in Result 1.

SELECT c.cid, name, oid, total FROM orders o LEFT OUTER JOIN customer c ON (c.cid = o.cid);

ii) The following codes produce the output in Result 2.
SELECT c.cid, name, oid, total FROM orders o LEFT OUTER JOIN customer c ON (c.cid = o.cid);

iii) The following codes produce the output in Result 1.
SELECT c.cid, name, oid, total FROM orders o INNER JOIN customer c ON (c.cid = o.cid);

iv) The following codes produce the output in Result 2.

SELECT c.cid, name, oid, total FROM orders o INNER JOIN customer c ON (c.cid = o.cid);

- [x] i) is correct.
- [x] iv) is correct.
- [ ] iii) is correct.
- [ ] ii) is correct.

Week 11 - Lecture - P51.

Question 21

2.5 / 2.5 pts

Which of the following statement(s) is(are) correct?

- [x] Apache Hive does not allow users to update or delete individual records.
- [ ] The general RDBMS and Apache Hive have data handling abilities at the petabyte and exabyte levels, respectively.
- [ ] Apache Hive supports transactions.
- [x] Apache Hive is capable of handling Big Data, but it suffers from high latency.

Week 11 - Lecture - P12.

Question 22

2.5 / 2.5 pts

Match the following tool with their most suitable use cases.

Flume

Gather logs from multiple sys

Kafka

Real-time event streaming an

Sqoop

Data transfer between Hadoo

Oozie

Running multiple jobs in paral

Week 8 lecture slides

Question 23

2.5 / 2.5 pts

Which of the following Apache Pig statement(s) is(are) correct?

i) infile = LOAD 'user' AS (name, phone);

ii) fs -put MyFile.txt src;

iii) fs -getmerge myreport/final.txt

iv) STORE result INTO 'finalres';

- [ ] Statement ii) puts the file "MyFile.txt" to the "src" folder on the local machine.
- [ ] Statement iv) save the content held in finalres to the variable called "result".
- [x] Statement i) loads data from the source called "user", and the data format is designed to be the name and the phone number for each instance.
- [x] Statement iii) merges data in the "myreport" directory and output the merged results in the file called "final.txt"

Week 09 - Lecture - P15.

Question 24

2.5 / 2.5 pts

Which of the following description(s) is(are) correct?

- [x] The main components of Apache Pig include the data flow language.
- [x] Grunt is the interactive shell where you can type in statements in Apache Pig.
- [ ] The main components of Apache Pig include the combiner.
- [x] The main components of Apache Pig include the interpreter and the execution engine.

Week 09 - Lecture - P6.

In the lab sessions, we imported several MySQL tables into Hive using Sqoop.

The imported tables are: **employees**, **customers**, **products**, **orders**, and **order_details**.

Please use Hive to answer the following questions:

Question 25

4 / 4 pts

Find the `cust_id` of the customer who spent the highest total amount in the city of 'Phoenix'.

1125373

```sql
SELECT c.city, c.cust_id, SUM(p.price) AS total_amount_spent
FROM customers c
JOIN orders o ON c.cust_id = o.cust_id
JOIN order_details od ON o.order_id = od.order_id
JOIN products p ON od.prod_id = p.prod_id
WHERE c.city = 'Phoenix'
GROUP BY c.city, c.cust_id
ORDER BY total_amount_spent DESC
LIMIT 5;
```

Query History Saved Queries Results (5)

	city	cust_id	total_amount_spent
1	Phoenix	1125373	531554
2	Phoenix	1164507	445792
3	Phoenix	1199084	363801
4	Phoenix	1030574	351709
5	Phoenix	1061710	339468

Incorrect

Question 26

0 / 4 pts

Find the `cust_id` of the customer who lives in the city of 'San Quentin' and purchased both 'Screen Cleaning Cloth (gray)' and 'Rechargeable Batteries (AA, 4 pack)'.

1020365, 1048678, 1151188

```sql
SELECT c.*
FROM customers c
LEFT SEMI JOIN (
    SELECT cust_id
    FROM orders o
    JOIN order_details od ON o.order_id = od.order_id
    WHERE od.prod_id = 1273719
) prod1 ON c.cust_id = prod1.cust_id
LEFT SEMI JOIN (
    SELECT cust_id
    FROM orders o
    JOIN order_details od ON o.order_id = od.order_id
    WHERE od.prod_id = 1273767
) prod2 ON c.cust_id = prod2.cust_id
WHERE c.city='San Quentin';
```

Query History Saved Queries Results (1)

	cust_id	fname	lname	address	city	state	zipcode
1	1151188	Raphael	Hamm	23489 North 14th Street	San Quentin	CA	94974

Question 27

4 / 4 pts

Find the highest total amount spent by a customer.

3543457

```sql
SELECT c.cust_id, SUM(p.price) AS total_amount_spent
FROM customers c
JOIN orders o ON c.cust_id = o.cust_id
JOIN order_details od ON o.order_id = od.order_id
JOIN products p ON od.prod_id = p.prod_id
GROUP BY c.cust_id
ORDER BY total_amount_spent DESC
LIMIT 5;
```

Query History Saved Queries Results (5)

	cust_id	total_amount_spent
1	1058286	3543457
2	1065880	3122579
3	1100357	3019363
4	1157510	2924523
5	1196473	2911816

Question 28

4 / 4 pts

Find the `cust_id` of the customer who purchased the most products in the city of 'Poland'.

1039690

```
SELECT c.city, c.cust_id, COUNT(od.prod_id) AS total_items_purchased
FROM customers c
JOIN orders o ON c.cust_id = o.cust_id
JOIN order_details od ON o.order_id = od.order_id
WHERE c.city = 'Poland'
GROUP BY c.city, c.cust_id
ORDER BY total_items_purchased DESC
LIMIT 5;
```

Query History Saved Queries Results (4)

	city	cust_id	total_items_purchased
1	Poland	1039690	20
2	Poland	1000008	15
3	Poland	1103166	6
4	Poland	1161698	3

Question 29

4 / 4 pts

Find the total amount spent by the customer who purchased the most items.

2849618

```
SELECT c.cust_id, SUM(p.price) AS total_amount_spent, COUNT(od.prod_id) AS total_items_purchased
FROM customers c
JOIN orders o ON c.cust_id = o.cust_id
JOIN order_details od ON o.order_id = od.order_id
JOIN products p ON od.prod_id = p.prod_id
GROUP BY c.cust_id
ORDER BY COUNT(od.prod_id) DESC, total_amount_spent DESC
LIMIT 5;
```

Query History Saved Queries Results (5)

	cust_id	total_amount_spent	total_items_purchased
1	1132475	2849618	312
2	1107669	2825111	299
3	1123033	2177055	295
4	1093072	2458506	294
5	1193649	2582321	289

Use MapReduce to extract the required information from the source data "access_log.gz" on the path "/home/training/training_materials/developer/data".

You should arrange the data in the following format: ***IP dd mmm yyyy***.

The ***IP*** represents the IP address, and the ***dd mmm yyyy*** is the standard format for the date.

An example of the source data before and after the process is given below:

- Original Source:
  10.223.157.186 - - [15/Jul/2009:14:58:59 -0700] "GET / HTTP/1.1" 403 202
- Processed data:
  10.223.157.186 15 Jul 2009

Load the processed data into Apache Pig.

Answer the questions below based on your data processing results.

Question 30

4 / 4 pts

Which IP address had the most connections logged in May?

10.56.48.40

```
(Apr,10.216.113.172,10656)
(Aug,10.216.113.172,12417)
(Dec,10.216.113.172,14745)
(Feb,10.216.113.172,13675)
(Jan,10.216.113.172,16504)
(Jul,10.216.113.172,20394)
(Jun,10.54.143.175,10328)
(Mar,10.216.113.172,14232)
(May,10.56.48.40,10731)
(Nov,10.216.113.172,10244)
(Oct,10.216.113.172,9508)
(Sep,10.216.113.172,11776)
```

```
-- Step 1: Use MapReduce to take the IP and the date, month, and year from the log file.
-- Step 2: Load the processed result into Pig and then do the analysis

inlog = LOAD 'OT3A3_res/part-r-00000' USING PigStorage('\t') AS (ip:chararray, date:chararray, month:chararray, year:int);
--DUMP inlog;

--filtered = FILTER inlog BY (month == 'Feb' OR month == 'May' OR month == 'Oct');

--grouped = GROUP filtered BY (ip, month);
grouped = GROUP inlog BY (ip, month);

--ip_cnt = FOREACH grouped GENERATE FLATTEN(group) AS (ip, month), COUNT(filtered) AS connection_count;
ip_cnt = FOREACH grouped GENERATE FLATTEN(group) AS (ip, month), COUNT(inlog) AS connection_count;

monthly_group = GROUP ip_cnt BY month;

max_counts_per_month = FOREACH monthly_group GENERATE group AS month, MAX(ip_cnt.connection_count) AS max_connection_count;

join_result = JOIN ip_cnt BY (month, connection_count), max_counts_per_month BY (month, max_connection_count);

final = FOREACH join_result GENERATE ip_cnt::month AS month, ip_cnt::ip AS ip, ip_cnt::connection_count AS connection_count;

DUMP final;
```

Question 31

4 / 4 pts

Which IP address has the most connections logged in August?

10.216.113.172


```
(Apr,10.216.113.172,10656)
(Aug,10.216.113.172,12417)
(Dec,10.216.113.172,14745)
(Feb,10.216.113.172,13675)
(Jan,10.216.113.172,16504)
(Jul,10.216.113.172,20394)
(Jun,10.54.143.175,10328)
(Mar,10.216.113.172,14232)
(May,10.56.48.40,10731)
(Nov,10.216.113.172,10244)
(Oct,10.216.113.172,9508)
(Sep,10.216.113.172,11776)
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--DUMP inlog;

--filtered = FILTER inlog BY (month == 'Feb' OR month == 'May' OR month == 'Oct');

--grouped = GROUP filtered BY (ip, month);
grouped = GROUP inlog BY (ip, month);

--ip_cnt = FOREACH grouped GENERATE FLATTEN(group) AS (ip, month), COUNT(filtered) AS connection_count;
ip_cnt = FOREACH grouped GENERATE FLATTEN(group) AS (ip, month), COUNT(inlog) AS connection_count;

monthly_group = GROUP ip_cnt BY month;

max_counts_per_month = FOREACH monthly_group GENERATE group AS month, MAX(ip_cnt.connection_count) AS max_connection_count;

join_result = JOIN ip_cnt BY (month, connection_count), max_counts_per_month BY (month, max_connection_count);

final = FOREACH join_result GENERATE ip_cnt::month AS month, ip_cnt::ip AS ip, ip_cnt::connection_count AS connection_count;

DUMP final;
```

Question 32

4 / 4 pts

What is the connection count of the IP address logged with the most connections in December?

14,745

```
(Apr,10.216.113.172,10656)
(Aug,10.216.113.172,12417)
(Dec,10.216.113.172,14745)
(Feb,10.216.113.172,13675)
(Jan,10.216.113.172,16504)
(Jul,10.216.113.172,20394)
(Jun,10.54.143.175,10328)
(Mar,10.216.113.172,14232)
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```


```
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monthly_group = GROUP ip_cnt BY month;

max_counts_per_month = FOREACH monthly_group GENERATE group AS month, MAX(ip_cnt.connection_count) AS max_connection_count;

join_result = JOIN ip_cnt BY (month, connection_count), max_counts_per_month BY (month, max_connection_count);

final = FOREACH join_result GENERATE ip_cnt::month AS month, ip_cnt::ip AS ip, ip_cnt::connection_count AS connection_count;

DUMP final;
```

Question 33

4 / 4 pts

What is the connection count of the IP address logged with the most connections in April?

10,656

```
(Apr,10.216.113.172,10656)
(Aug,10.216.113.172,12417)
(Dec,10.216.113.172,14745)
(Feb,10.216.113.172,13675)
(Jan,10.216.113.172,16504)
(Jul,10.216.113.172,20394)
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```

```
-- Step 1: Use MapReduce to take the IP and the date, month, and year from the log file.
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join_result = JOIN ip_cnt BY (month, connection_count), max_counts_per_month BY (month, max_connection_count);

final = FOREACH join_result GENERATE ip_cnt::month AS month, ip_cnt::ip AS ip, ip_cnt::connection_count AS connection_count;

DUMP final;
```

Question 34

4 / 4 pts

What is the connection count of the IP address logged with the most connections in February?

13,675

```
(Apr,10.216.113.172,10656)
(Aug,10.216.113.172,12417)
(Dec,10.216.113.172,14745)
(Feb,10.216.113.172,13675)
(Jan,10.216.113.172,16504)
(Jul,10.216.113.172,20394)
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(Mar,10.216.113.172,14232)
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join_result = JOIN ip_cnt BY (month, connection_count), max_counts_per_month BY (month, max_connection_count);

final = FOREACH join_result GENERATE ip_cnt::month AS month, ip_cnt::ip AS ip, ip_cnt::connection_count AS connection_count;

DUMP final;
```

Quiz score: 88.92 out of 100